PROJECT EXPLANATION-2.O

Hi, my name is Mustafa. I'm a DevOps Engineer with around 4 years of hands-on experience designing and implementing CI/CD pipelines, containerized applications, and cloud-native infrastructure. My core strengths are in Jenkins, Docker, Kubernetes, Terraform, and implementing DevSecOps practices using tools like Trivy, SonarQube, TFSec, and OWASP ZAP.

I've worked closely with development, QA, and cloud engineering teams to automate build, deployment, and infrastructure provisioning. I enjoy improving reliability, reducing deployment time, and strengthening security in the delivery process.

Project Overview
I worked on a microservices-based application where my main responsibility was to automate the CI/CD pipelines, containerize services, and manage deployments on Kubernetes using infrastructure as code principles.

2. My Responsibilities
CI/CD Pipeline Automation
Designed and implemented multi-stage Jenkins pipelines (Build → Test → Scan → Deploy).
Integrated SonarQube for code quality and OWASP ZAP for dynamic security scanning.
Added Trivy image scanning steps in the pipeline to block vulnerable images.
Automated unit tests, code coverage, integration tests, and blue-green deployments.

Containerization & Kubernetes

Containerized applications using Docker following best practices (small images, multi-stage builds).
Deployed microservices on Kubernetes clusters.
Managed scaling, rolling updates, resource limits, and readiness/liveness probes.

Infrastructure as Code (Terraform)
Provisioned AWS (or other cloud) infrastructure using Terraform (VPC, EKS, RDS, IAM, ALB).
Implemented TFSec to validate Terraform code for security vulnerabilities.

DevSecOps
Integrated Trivy, SonarQube, TFSec, OWASP ZAP into CI/CD.
Ensured compliance with security standards (CIS benchmarks, least-privilege roles).

Monitoring & Logging
Set up Prometheus + Grafana dashboards for cluster & service metrics.
Configured ELK/EFK stack for centralized logging.

Day-to-Day Activities:

1. Monitoring & Support
Review Jenkins jobs, Kubernetes pods, cluster health.
Investigate deployment failures or performance issues.
Respond to alerts from Prometheus/Grafana.

2. CI/CD Pipeline Development
Create or enhance Jenkins pipelines.
Add testing, security scanning, and automated rollback steps.
Convert manual deployment tasks into automated scripts.

3. Kubernetes Operations
Manage deployments, StatefulSets, Ingress, ConfigMaps, Secrets.
Optimize resource requests & limits.
Troubleshoot pod crashes, networking issues, or service downtime.

4. Infrastructure as Code (Terraform)
Create/modify Terraform modules for new infrastructure requirements.
Run TFSec checks and plan/apply changes through PR approvals.

5. Security & Compliance
Run Trivy image scans.
Review SonarQube quality gates.
Ensure OWASP ZAP results are addressed by dev teams.
Patch container base images.

6. Collaboration
Discuss changes with developers, QA, cloud, and security teams.
Suggest optimizations to improve deployment speed and stability.

7. Documentation & Continuous Improvement
Update architecture diagrams, runbooks, and deployment guides.
Propose improvements to CI/CD, Kubernetes, and cloud infrastructure.